01. TCP Three-Way Handshake, Teardown and TCP Flags

1. TCP

TCP (Transmission Control Protocol) is a connection-oriented protocol that provides reliable data transmission between two devices over a network.

Key Points

- TCP establishes a connection before transferring data.
- It ensures data arrives in the correct order.
- It detects lost data and retransmits it when necessary.
- It uses acknowledgements (ACKs) to confirm received data.
- It is used by protocols such as HTTP, HTTPS, SSH and FTP.

Example: When you open a website, TCP may establish a connection between your computer and the web server before HTTP data is exchanged.

2. TCP Three-Way Handshake

The TCP three-way handshake is the process used to establish a TCP connection between a client and a server.

Three Steps

Step 1: SYN (Synchronize)

- The client sends a SYN packet to the server.
- It requests a TCP connection.
- It includes an initial sequence number.

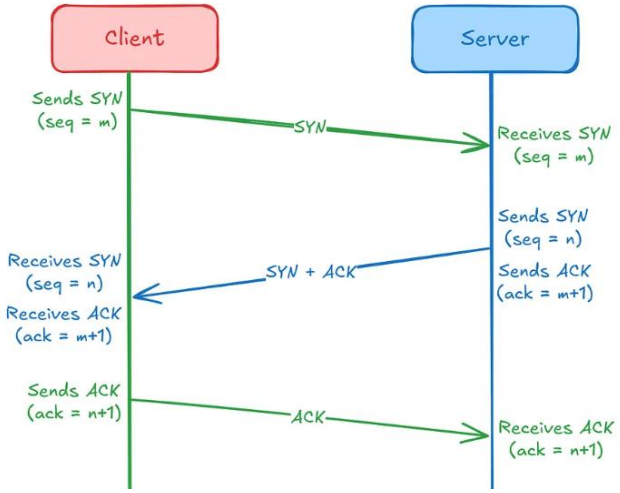
Step 2: SYN-ACK (Synchronize-Acknowledge)

- The server replies with SYN and ACK flags set.
- SYN indicates that the server is synchronizing its sequence number.
- ACK acknowledges the client's SYN.

Step 3: ACK (Acknowledge)

- The client sends an ACK packet.
- It acknowledges the server's SYN.
- The TCP connection is established, allowing data transfer.

Handshake Diagram



Handshake Table

Step	Packet	Purpose
1	SYN	Client requests a connection
2	SYN-ACK	Server accepts and acknowledges the request
3	ACK	Client acknowledges the server's response

Why Is the Handshake Important?

- It establishes a TCP connection.
- Both devices synchronize sequence numbers.
- It helps analysts identify connection attempts.
- Repeated SYN packets without completed handshakes may indicate scanning or a SYN flood, but further investigation is required.

3. TCP Connection Teardown

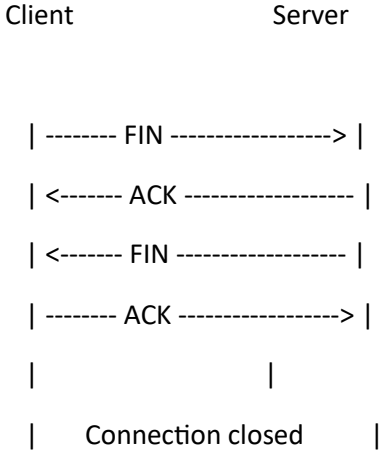
TCP teardown is the process of closing an established TCP connection.

A normal TCP connection is commonly closed using FIN and ACK packets.

Four-Step Teardown

1. The client sends FIN to indicate that it has finished sending data.
2. The server sends ACK to acknowledge the FIN.
3. The server sends FIN when it has finished sending data.
4. The client sends ACK to acknowledge the server's FIN.

Teardown Diagram



Note: TCP may close a connection differently in some situations. For example, a RST packet can terminate a connection abruptly.

4. TCP Flags

Definition: TCP flags are control bits in the TCP header that help manage connection establishment, data transfer and connection termination.

Flag	Full Form	Purpose
SYN	Synchronize	Starts a connection and synchronizes sequence numbers
ACK	Acknowledgment	Acknowledges received data or connection messages
FIN	Finish	Indicates that a device has finished sending data
RST	Reset	Resets or abruptly terminates a connection
PSH	Push	Requests that received data be passed to the application promptly
URG	Urgent	Indicates that the urgent pointer field is significant

Examples of TCP Flags

- **SYN:** A client attempts to connect to a web server.
- **ACK:** A device acknowledges a received TCP segment.
- **FIN:** A client closes a normal TCP connection.
- **RST:** A server refuses a connection or resets an existing connection.
- **PSH:** A TCP segment carries application data that should be delivered promptly.
- **URG:** A segment marks urgent data using the TCP urgent pointer; uncommon in most modern applications.

5. TCP Analysis in Wireshark

Wireshark is a network protocol analyzer used to capture and inspect packets.

Useful Display Filters

Filter	Purpose
tcp	Shows TCP packets
tcp.flags.syn == 1	Shows packets with SYN set
tcp.flags.syn == 1 && tcp.flags.ack == 0	Shows initial SYN packets without ACK
tcp.flags.syn == 1 && tcp.flags.ack == 1	Shows SYN-ACK packets
tcp.flags.fin == 1	Shows packets with FIN set
tcp.flags.reset == 1	Shows packets with RST set

What Should a SOC Analyst Look For?

- Many SYN packets sent to different ports in a short period.
- Repeated SYN packets without a corresponding SYN-ACK or completed handshake.
- Unexpected RST packets or repeated connection failures.
- Connections to unusual destination ports.
- Unusual connection patterns from unknown devices.

Important: These patterns do not automatically prove an attack. An analyst should check the source IP, destination IP, timestamps, ports and surrounding traffic before deciding whether activity is suspicious.

6. Conclusion

TCP provides reliable communication using sequence numbers, acknowledgements and control flags. The three-way handshake establishes a connection, while FIN and ACK packets commonly support normal teardown. Wireshark allows analysts to inspect these packets and identify unusual connection patterns that may require further investigation.